

DASR Supplement 2: Theory Statements and Proofs

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1 Registered-batch execution

An occupation-measure solution can contain fractional action shares at one state. The registered executor realizes such a row within an evaluation batch by calibrated-score ordering, deterministic integer allocation, and stable tie-breaking. We write π_D for the resulting deterministic execution on registered batch D . The quotient, comparison, and edit statements use this registered execution unit.

2 Finite audit grouping

For n stopped-negative cases and zero observed excess events, a one-sided exact upper confidence bound at error level α is $1 - \alpha^{1/n}$.

Proposition 1 (Support threshold for a zero-event cell). *To make this upper bound at most γ , one needs*

$$n \geq \frac{\log \alpha}{\log(1 - \gamma)}.$$

At the registered per-slice level $\alpha = 0.0025$, this gives $n \geq 117$ for $\gamma = 0.05$ and $n \geq 131$ for $\gamma = 0.045$.

Proof. Solve $1 - \alpha^{1/n} \leq \gamma$ for n , observing that both logarithms are negative. The numerical values follow by taking ceilings. $\square$

This proposition supports the predeclared pooled audit vocabulary. Decision value governs the searched routing states below.

3 Balanced proposals and minimal release

Proposition 2 (Equal-mass binary proposal). *For a parent with fixed construction count n split into counts $n_1 + n_2 = n$, the larger child mean-contrast variance factor $\max\{1/n_1, 1/n_2\}$ is minimized when the counts are as equal as integrality permits.*

Proof. If $n_1 < n_2 - 1$, moving one unit from child 2 to child 1 strictly reduces the maximum reciprocal. Repeating yields $|n_1 - n_2| \leq 1$. $\square$

Lemma 1 (Minimal child/complement release). *Let C be a registered child of pooled state B . The partition $\{C, B \setminus C\}$ is the coarsest refinement under which C can receive an action different from the remainder, and it adds one independent control contrast.*

Proof. Any representation assigning a separate action to C must distinguish membership in C , hence refine the two-cell partition. Conversely the two cells suffice. Two cell actions relative to one pooled action create one free difference. $\square$

4 Fixed-solve action-face value

Fix a parent B at a solved policy. Let its pooled action be a_B . Descendants k have occupation weights w_k summing to one. For alternative b , define $d_k(b) = Q_k(b) - Q_k(a_B)$, mean $g_B(b) = \sum_k w_k d_k(b) > 0$, and centered mean absolute dispersion $H_Q(B, b) = \sum_k w_k |d_k(b) - g_B(b)|$.

Theorem 1 (Crossing mass and released value). *The occupied mass of descendants satisfying $d_k(b) < 0$ is at most $H_Q(B, b)/(2g_B(b))$. Summed across alternative actions, released action value per unit parent mass is at most $\frac{1}{2} \sum_{b \neq a_B} H_Q(B, b)$.*

Proof. Write $u_k = d_k(b) - g_B(b)$, so $\sum_k w_k u_k = 0$. A crossing requires $u_k < -g_B(b)$. Therefore

$$g_B(b) \sum_{u_k < -g_B(b)} w_k \leq \sum_{u_k < 0} w_k (-u_k).$$

For a centered random variable, total positive and negative deviations coincide and each equals half the mean absolute deviation. Thus the right side is $H_Q(B, b)/2$. The same negative-deviation identity bounds the improvement obtained by switching crossing descendants, and summing over alternatives gives the second claim. $\square$

The theorem ties resolution to Bellman action contrasts. Certified routing value arises from distinctions near an action face.

5 CVaR resolution charge

Fix public slice weights, descendant slice risks, CVaR hinge η , and tail-row dual price ν . Let $h(q) = \{q - \eta\}_+$.

Theorem 2 (Resolution exposes nonnegative hinge mass). *For each parent and public slice, Jensen's inequality gives*

$$\sum_k w_k h(q_k) - h\left(\sum_k w_k q_k\right) \geq 0.$$

Let D_B be the sum of these discrepancies with the registered tail normalization. If V_B is the fixed-solve action-face value and μ_B the parent occupation mass, then the fixed-solve refinement ledger is

$$\Delta_B = V_B - \nu \mu_B D_B.$$

$D_B = 0$ exactly when every occupied descendant within each public slice lies on one side of the hinge (allowing equality at the hinge).

Proof. The hinge is convex, which proves nonnegativity. Equality for a piecewise-linear hinge occurs when the mixture support remains in one affine region. In the fixed Lagrangian, action improvement and the newly exposed tail row enter additively, with multiplier ν and parent mass μ_B . $\square$

The identity is local to a fixed solve. DASR follows accepted admissions with a constrained master-program re-solve.

6 Frozen-frontier properties

Proposition 3 (Registered-batch executable-table quotient). *If two frozen policies define the same state-address rules and action fractions at every row, the same registered executor applied to the same batch produces identical evidence routes and terminal actions. Their observed costs, slice counts, audit statistic, and paired comparisons on that batch are identical. Removing exact duplicate executable tables is therefore lossless under the registered execution specification.*

Proof. The two tables present identical shares to the same ordering, allocation, and tie-breaking rules at every reached row. Induction over tiers gives identical batch routes and actions. Every recorded aggregate is a deterministic function of those executions, outcomes, and the fixed operating specification. $\square$

Proposition 4 (Selection-split menu monotonicity). *For a frontier frozen before Policy Selection, with a pointwise screen independent of family size and least-cost selection among survivors, adding a predeclared candidate preserves or lowers selected Policy Selection cost.*

Proof. The old survivor set remains a subset of the enlarged survivor set. The minimum over the enlarged set is at most the original minimum. $\square$

Definition 1 (Disagreement-localized registered-batch radius). *For two frozen registered-batch executions, let X_i be their observed per-unit cost difference bounded in absolute value by L , let $\widehat{V}$ be its empirical variance on the common batch, and let c be a fixed empirical-Bernstein constant. We record the descriptive radius*

$$\sqrt{\frac{2\widehat{V} \log(2M^2/\delta)}{n}} + \frac{cL \log(2M^2/\delta)}{n}.$$

Since $X_i = 0$ on route-agreement cases, $\widehat{V}$ contracts with observed route-disagreement mass.

For a casewise policy or an independently randomized per-case kernel, the usual independence conditions support a population empirical-Bernstein theorem and a union bound over policy pairs. The registered executor records these radii as fixed-batch paired diagnostics. The selected policy’s independent audit supplies the safety result.

7 Operator geometry

Theorem 3 (Exact repricing and vertex regret). *Let $v(\pi)$ contain the three tier-reach probabilities, false-positive mass, and review mass. Then $J_\theta(\pi) = \theta^\top v(\pi)$ for every price vector θ in the declared product domain. Against a finite saved library, a policy’s maximum regret over the domain is attained at one of its nine vertices.*

Proof. The five consequences exhaust every price-dependent term in realized cost, giving the dot product. Regret against a fixed comparator is linear in θ ; regret against the best saved comparator is the maximum of finitely many linear functions, hence convex and piecewise linear. A convex function over a polytope attains its maximum at a vertex. $\square$

Proposition 5 (Fail-closed address-edit verification). *For a registered batch and executor, a direct edit at address C is accepted only after replay verifies that every changed route belongs to the declared trace-hit set of C . The complete edited execution must then pass the global Policy Selection screen and, if forwarded, the independent Sealed Audit.*

Proof. The implementation compares pre-edit and post-edit routes on the complete registered batch and verifies that every changed route is within the declared address set. It then applies the complete-policy screen and audit. This supplies an exact procedural guarantee for the replayed batch. $\square$